

Modified Full Bayesian Networks Classifiers for Medical Diagnosis

Ahmed T. Sadiq AlObaidi
Department of Computer Sciences
University of Technology
Baghdad, Iraq
e-mail: drahmaed_tark@yahoo.com

Noor Thamer Mahmood
Department of Computer Sciences
Baghdad University
Baghdad, Iraq
e-mail: noorthamer@gmail.com

Abstract— This paper is present a modified Bayesian Network (BN) for Full Bayesian Classifier (FBC) for used it to diagnoses the condition of a patient from some of symptoms and medical history. The conditions are the heart diseases and the nervous diseases. Modified for FBC is depended on common structure known as *naïve Bayes*. Determining the network structure is D-separated by the variable. Each variable has CPT and each disease (table) has Probability. By modified the equation for find CPT for each variable and Probability for each disease (table) in Modified-FBC (M-FBC) structure, the system for diagnosis is designed. The experimental resulted show that the successful ratio of heart diseases (93%) and nervous system diseases (98%) approximately.

Keywords- Bayesian Network (BN); Full Bayesian Classifier (FBC); Modified-FBC (M-FBC); CPT; Probability; *naïve Bayes*

I. INTRODUCTION

A Bayesian network consists of two parts: the structure and the parameters. The structure is represented by a directed acyclic graph (DAG) and the parameters determine the conditional probability distributions for each node [1].

A DAG is a pair (N, A) , where N is the node set and $A \subseteq N \times N$ is the arc set. Note that a DAG is acyclic, that is, the arcs in A do not form directed cycles. A node u is said to be a parent of a node v if the arc set contains an arc from u to v , that is, $uv \in A$. The set of the parents of v are denoted by Av . If u is a parent of v then v is a child of u . Further, a node u is said to be an ancestor of a node v in a DAG (N, A) if there is a directed path from u to v in A ; denoted $u \rightarrow^* v$. If u is an ancestor of v , then v is a descendant of u . When there is no ambiguity about the node set, we identify a DAG by its arc set. The cardinality of N is denoted by n [1].

The method in this paper using the Bayesian Networks (BNs) to extract the structure that classify a medical data. That's where these conditions attached are cases of medical data as possible to take advantage of them in the formation of the structure of BNs. Causal and Evidential are most common types of reasoning in BNs, however thanks to evidential reasoning Bayesian networks became the ubiquitous tool for differential diagnosis even though this type requires much more computational power than causal reasoning. On the other hand causality is the fundamental

part of Bayesian networks (BNs) and without it there would be no reasoning [2].

Database consists of two types of data for the heart and the nervous system diseases, where taken five types for each diseases data, these diseases have 20 of the common cases among patients for each one. Through the input stage of the symptoms and the patient's medical history is being processed through the BNs structure resulting from FBN Modified. By modified the equation for find CPT and Probability for each variable in Modified-FBC (M-FBC) structure, the system for diagnosis is designed. This way gets the exact diagnosis of a patient's disease.

Aim of paper is speed of predict for disease infected by the patient in order to save his life. By explaining the patient's symptoms experienced by mentions for medical history is through the introduction of all the information into the system to reach the approximate diagnosis of a patient's disease patient.

The system has been applied in Iraqi hospitals and reached acceptable results and that a certificate therapists doctors in the hospital.

Where the margin of error were identified by doctors around 5% for most of the cases.

So that it is accurate diagnosis for heart and the nervous system diseases, thus can enter other data set for diseases is entered into the system is up the diagnostic process as well.

Thus, we conclude that the use of the BNs structure get to make a decision regarding the diagnosis through the work of deductive relations between the attributes to get to the appropriate structure of the disease and the disease is diagnosed through this structure of data-based truth until we get to the best results.

II. RELATED WORK

Beáta Reiz, Lehel Csató, 2009[3]. Bayesian networks encode causal relations between variables using probability and graph theory. They can be used both for prediction of an outcome and interpretation of predictions based on the encoded causal relations. In this paper we analyze a tree-like Bayesian network learning algorithm optimized for classification of data and we give solutions to the interpretation and analysis of predictions. The classification of logical – i.e. binary – data arises specifically in the field of medical diagnosis, where we have to predict the survival chance based on different types of

medical observations or we must select the most relevant cause corresponding again to a given patient record. Surgery survival prediction was examined with the algorithm. Bypass surgery survival chance must be computed for a given patient, having a data-set of 66 medical examinations for 313 patients.

Li Hongyan, Wu Zhong, Li Cheng, 2009 [4]. This article combines with China's aviation status, conducts a more deep and systematic study of risk analysis and decision-making in overbooking strategy through using qualitative and quantitative method, which contributes a certain practical value. Overbooking refers that an airlines allow confirmed booking beyond the seating capacity for the flight based on reasonable estimation. The airlines are allowed a certain amount of overbooking because cancellations are fairly frequent for many reasons. There are no cancellation charges, only paying a few indemnities even when returning tickets or missing flights after purchasing tickets. Overbooking surely runs a certain risk. Usually, overbooking is estimated by cancellation of reservation and probability of passenger's no-show. Thus, there may be excessive overbooking and the occurrence of super-real, some passengers cannot check-in, thus, extra compensation, operating loss as well as loss of goodwill are created for airlines. Therefore, it is obviously necessary for overbooking to reduce possibilities of vacancy losses. The overbooking is more, the possibility of vacancy lose is smaller. Overbooking is an effective means for airlines to enhance revenue. Today, overbooking has been commonly used by airlines worldwide and already achieved a certain economic benefits.

Silvia Acid, Luis M. de Campos, Juan M. Fernandez-Luna, Susana Rodriguez, Jose' Maria Rodriguez, Jose' Luis Salcedo, 2004 [5]. Due to the uncertainty of many of the factors that influence the performance of an emergency medical service, we propose using Bayesian networks to model this kind of system. We use different algorithms for learning Bayesian networks in order to build several models, from the hospital manager's point of view, and apply them to the specific case of the emergency service of a Spanish hospital. This first study of a real problem includes preliminary data processing, the experiments carried out, the comparison of the algorithms from different perspectives, and some potential uses of Bayesian networks for management problems in the health service.

Stefan Conrady, Dr. Lionel Jouffe, 2011 [6]. By using Bayesian networks as the framework, we have shown a practical new modeling approach based on the widely studied Wisconsin Breast Cancer Database. Our prediction accuracy is comparable with the results of all known studies on this topic. With BayesiaLab as the software tool, modeling with Bayesian networks becomes accessible to a very broad range of analysts and researchers, including non-statisticians. The speed of modeling, analysis and subsequent implementation make BayesiaLab a suitable tool

in many areas of research and especially for translational science.

Jyotirmay Gadewadikar, Ognjen Kuljaca, Kwabena Agyepong, Erol Sarigul, Yufeng Zheng and Ping Zhang, 2010 [7]. The objective of this paper is to explore the implementation of a Bayesian Belief Network for an automated breast cancer detection support tool. It is intuitive that Bayesian networks are employed as one viable option for computer-aided detection by representing the relationships between diagnoses, physical findings, laboratory test results and imaging study findings. This work brings important entities such as Radiologists, Image Processing Scientists, Data Base Specialists and Applied Mathematicians on a common platform. A brief background concerning causal networks, probability theory and Bayesian networks is given; available computational tools and platforms are described. It is explained that, by exploiting conditional independencies entailed by influence chains, it is possible to represent a large instance in a Bayesian network using little space, and it is often possible to perform probabilistic inference among the features in an acceptable amount of time. The next steps towards realizing a Bayesian Belief Network Implementation are described. Bayesian networks have an unparalleled advantage of being able to exploit the explicit structure of the domain model to derive a graphical representation for learning. The encoding of independencies in the network topology admits the design of efficient procedures for performing computations over the network. For the application of computer-aided detection in mammography, the researchers intend to design an interface between the project's Bayesian network learning algorithm and the radiologists, so that the radiologists can have interaction with the system by labeling only a small number of informative images presented by the active learning algorithm.

III. BAYESIAN NETWORKS

Bayesian networks (BNs) are triplets (V, E, P) , where (V, E) is a directed acyclic graph (DAG) with nodes V , edges E , and a set of probability distributions P , called parameters, whose elements are assigned to the nodes of the graph. The nodes represent domain variables and edges mark direct causal relations between these variables [3].

The network encodes a joint probability distribution function representative to the domain:

$$P(\mathcal{X}) = \prod_{i=1}^n P(X_i | \text{par}(X_i)) \quad (1)$$

where n is the number of domain variables, X_i is a node from the BN and $\text{par}(X_i)$ is the set of X_i 's parents. The probability of the graph ensures the product to be finite [3].

We employed a tree-like representation for the topology of BN in order to increase efficiency in class variable estimation and interpretation. We describe this algorithm, where we construct a tree in such a way that the root of the tree will be the class variable and the remaining nodes are attributes [3].

Direct causal relations encoded by the BN are interpreted as the maximum of mutual respective conditional mutual information between nodes. Now we present the necessary information theoretical concepts for our algorithm [3].

We will use the following notations: X and Y are random variables defined on probability spaces Ω_X respective Ω_Y with corresponding distribution functions $p(x)$ respective $p(y)$. We use their joint and conditional probability functions, denoted with $p(x,y)$ and $p(x|y)$ respectively[3].

Mutual information is the quantity of information two random variables contain about each other, defined as [8]:

$$I(X,Y) = \sum_{x \in \Omega_X} \sum_{y \in \Omega_Y} p(x,y) \log_2 \frac{p(x,y)}{p(x)p(y)} \quad (2)$$

In next sections we present a Bayesian network structure learning algorithm. The algorithm consists of an extension of Bayesian structure learning algorithm with finding causal and evidential reasoning relations between attributes.

IV. LEARNING FBC

Given a training data set S , we partition S into $|C|$ subsets, each S_c of which corresponds to the class value c , and then construct an *FBC* for S_c . Note that learning the structure of a full BN actually means learning an order of variables and then adding arcs from a variable to all the variables ranked after it. In fact, given the order of variables, learning a BN is relatively easy. However, learning the order of variables is still difficult. Recently, Teyssier *et al.* proposed to learn the order of variables through a sophisticated local search [8]. We adopt a straightforward method, which ranks a variable based on its total influence on other variables. The influence (dependency) between two variables can be measured by mutual information defined as follows [9].

$$M(X,Y) = \sum_{x,y} p(x,y) \log_2 \frac{p(x,y)}{p(x)p(y)}, \quad (3)$$

where x and y are the values of variables X and Y respectively. Note that, since we compute the mutual information in each subset S_c of the training set, $M(X,Y)$ is

actually the conditional mutual information $M(X,Y|c)$. In practice, it is possible that the dependency between two variables, measured by Equation 3, is caused merely by noise. In that case, the dependency is not reliable and should not be taken into account. Thus, we need a threshold to judge whether the dependency between two variables is reliable. One typical approach to defining the threshold is based on the Minimum Description Length (MDL) principle. Friedman *et al.* show that the average cross-entropy error is asymptotically proportional to, where m is the size of the training data [10]. We adopt their result to define the threshold to filter out unreachable dependencies as follows [9].

$$\varphi(X_i, X_j) = \frac{\log(m)}{2m} \times T_{ij}, \quad (4)$$

where $T_{ij} = |X_i| \times |X_j|$, $|X_i|$ is the number of possible values of X_i , and $|X_j|$ is similar. Assume that we use the most naive form as the representation for CPTs. Then, if X_i and X_j are independent, the total size of two CPTs for X_i and X_j is $|X_i| + |X_j|$. If we represent the dependency between X_i and X_j , the total size of the two CPTs is $|X_i| \times |X_j| + |X_i|$ (Assuming X_i is the parent of X_j). Thus, roughly speaking, T_{ij} represents the size increase for representing the dependency between X_i and X_j . In our implementation, the dependency between X_i and X_j is taken into account only if $M(X_i, X_j) > \varphi(X_i, X_j)$ [9].

The total influence of a variable X_i on other variables, denoted by $W(X_i)$, is defined as follows.

$$W(X_i) = \sum_{X_j \in C, X_j \neq X_i}^{M(X_i, X_j) > \varphi(X_i, X_j)} M(X_i, X_j). \quad (5)$$

Consequently, we have the following learning algorithm based on $W(X_i)$ [4].

Algorithm *FBC-Structure*(S, X)

Input: A set S of labeled, a set X of variables.

Output: A full BN B .

1. $B = \text{empty}$.
2. Partition the training data S into $|C|$ subsets S_c by the class value c .
3. **For** each training data set S_c

- Compute the mutual information $M(X_i, X_j)$ and the structure penalty $\phi(X_i, X_j)$ between each pair of variables X_i and X_j based on Equation 3 and Equation 4.
- Compute $W(X_i)$ for each variable X_i based on Equation 5.
- For all variables X_i in X
 - Add all the variables X_j with $W(X_j) > W(X_i)$ to the parent set Π_{X_i} of X_i .
 - Add arcs from all the variables X_j in Π_{X_i} of X_i .
- add the resulting network B_c to B

4. Return B .

V. LEARNING FAST-CPT-TREES

After the structure of an FBC is determined, a CPT-tree should be learned for each variable X_i . Certainly, a traditional decision tree learning algorithm, such as C4.5, can be used to learn Fast-CPT-trees. Note that a decision tree should be learned for each variable. A fast decision tree learning algorithm for learning CPTs. Before the tree-growing process, all the variables X_j in Π_{X_i} are sorted in terms of mutual information $M(X_i, X_j)$ on the whole training data, which determines a fixed order of variables. In the tree growing process, remove the variable X_j with the highest mutual information from Π_{X_i} , and compute the local mutual information $M^S(X_i, X_j)$ on the current training data S . If it is greater than the local threshold $\phi^S(X_i, X_j)$, X_j is used as the root, and partition the current training data S according to the values of X_j and repeat this process for each branch (subset). Otherwise, try another variable. The key point here is that, for each variable, we compute the local mutual information and local threshold just once. Regardless of whether the attribute chosen is failed or not, it is removed from Π_{X_i} and is never considered again. In contrast, the traditional decision tree learning algorithm computes the local mutual information of each unused variable at each step. The algorithm is depicted as follows [9].

Algorithm Fast-CPT-Tree (Π_{X_i}, S)

Input: Π_{X_i} is the set of parents of X_i , and S is a set of labeled examples.

Output: A decision tree T with X_i as the predictive variable.

1. Create an empty tree T .
2. If (S is pure or empty) or (Π_{X_i} is empty) Return T .
3. $qualified = False$.
4. While ($qualified == False$) and (Π_{X_i} is not empty)
 - Choose the variables X_j with the highest $M(X_i, X_j)$.
 - Remove X_j from Π_{X_i} .
 - Compute the local mutual information $M^S(X_i, X_j)$ on S based on Equation 3.
 - Compute the local structure penalty if $\phi^S(X_i, X_j)$ on S based on Equation 4.
 - If $M^S(X_i, X_j) > \phi^S(X_i, X_j)$ $qualified = True$
5. If $qualified == True$
 - Create a root X_j for T .
 - Partition S into disjoint subsets S_x , where x is a value of X_j .
 - For all values x of X_j
 - $T_x = \text{Fast-CPT-Tree}(\Pi_{X_i}, S_x)$
 - Add T_x as a child of X_j .
6. Return T .

VI. THE PROPOSED MEDICAL ESTIMATION SYSTEM(MES)

The propose system consists several phasaes of the application of the system.

Phase 1: Is to acquire data were obtained from the patients' files in Iraqi hospitals. We entered the data acquired to the system, which are 10 diseases are divided to 5 for heart diseases and 5 for nervous system diseases.

Phase 2: In this phase is filtering and extracting data for taking only the main symptoms and medical history for each disease. Account the patients have the same symptoms and medical history in the new table in database. Each table is stored faithful disease cases of the disease and recurrence of symptoms similar cases are enumerated. The development of these symptoms Similar on-line in a separate table of the disease so that 20 is configured under different symptoms for a number of patients with these symptoms for the same disease. This was configured 10 tables, each table represents a particular disease five tables represent heart disease and the other five represent Diseases of the nervous. As for the

fields each table consists of a number of different attributes from the other table. There is a Table (1, 2) explains all the tables of every detail.

Phase 3: After filtering and extracting data store all tables in repository of feature DB show Table (1, 2) are used in phase 4.

TABLE I. DETAIL FOR HEART DISEASES

Code Diseases	Diseases Name	Records Number	Attributes Number	Patients Number
I10	High Blood Pressure	20	19	200
I20	Angina pectoris	20	26	330
I21	Myocardial Infarction	20	24	450
I48	Atrial Fibrillation	20	14	390
I50	Heart Failure	20	28	565

TABLE II. DETAIL FOR NERVOUS SYSTEM DISEASES

Code Diseases	Diseases Name	Records Number	Attributes Number	Patients Number
G40	Epilepsy	20	15	268
G55	Nerve Root and Plexus Compressions	20	11	238
G91	Hydrocephalus	20	14	188
G94	Head injury	20	18	190
G95	Other diseases of spinal cord	20	14	228

Where, No. Of Disease: Statistical represents the number of disease.

Phase 4: Input all database to models learning bayesian networks all models are taken the database and make the process for all database. In *M-FBC* Model is depending on calculat (mutual information , threshold, wieght for each variable, modified CPT for each variable and modified Probabilty for structure).

Phase 5: After the process leads to the many models and reasoning BNs (structure BNs). Choose the structure output of *M-FBC* Model see section (6.1).

Phase 6: Taking the structure *M-FBC* and making evaluation BNs fitness by using some test data for testing the structures select and save only fitness structure BNs.

See Figure (1) to see all Phases in block diagram for Medical Estimation System (MES)

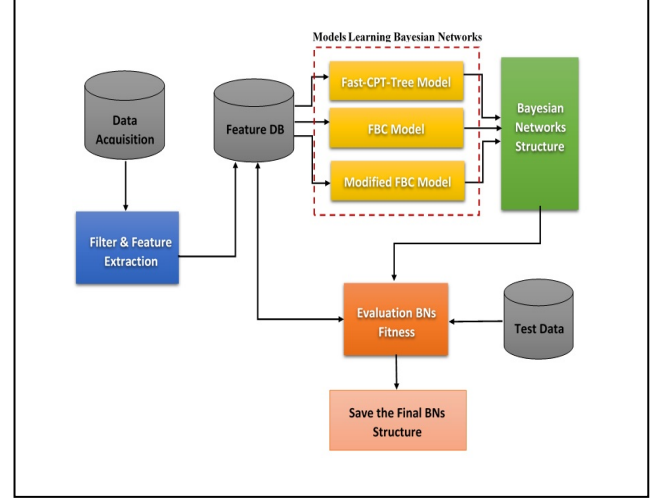


Figure 1. Block Diagram MES.

VI.1 LEARNING MODIFIED-FBC (*M-FBC*)

After the phase Filter and Feature Extraction the dataset is stored in the repository Feature DB. All dataset becomes input to *M-FBC* model. Checked the table in DB is empty or not and calculate the mutual information for each pair of attributes in the table base on equation (3) for purpose of knowledge as a measure of relationship between two random attributes that are in table simultaneously. After that calculate a threshold between each pair of attributes base on equation (4) and selected the maximum threshold between attributes. Because the equation (3) caused merely by noise, thus we need a threshold to rule whether the dependency between to variables is reliable.

We need to calculate weight for each attribute in the table base on equation:

$$W(x_i) = \frac{MI(x_i, Y) - \max_{j \neq i} MI(x_j, Y)}{\sum_{j \neq i} MI(x_j, Y)} \quad (9)$$

Take all weight for variables and make descending order and select the maximum is become a root and the other is evidence for root show Figure (2) the structure is call *naïve Bayes* but Inverted.

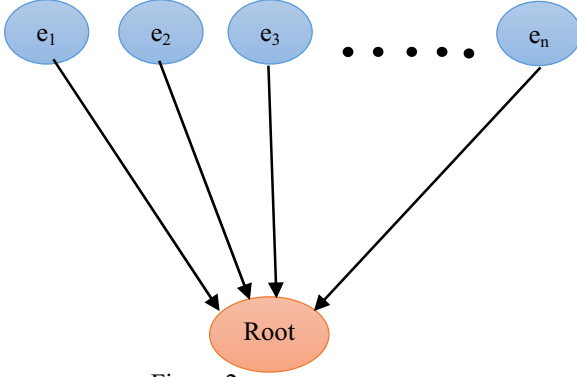


Figure 2: common effect.

Calculate CPTs for each node in network structure is D-separate see Figure (3) either node is positive then selected when node=1 is (sum no. of node which value =1 / total node=1 and node=0) or node is negative then selected when node=0 is 1- (sum no. of node which value =0 / total node=1 and node=0).

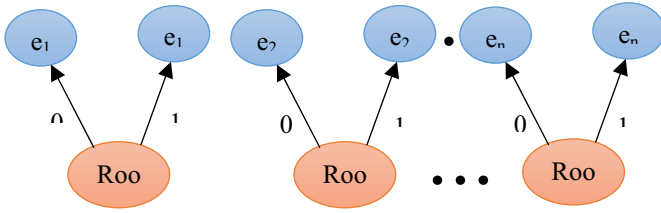


Figure 3: CPTs (common causal).

Algorithm *M-FBC Structure(S, X)*

Input: Table (i), (i=1,..., n) .

Output: A full BNs **B**, Probability (Prob).

1. B = empty.
2. If Table (i) is empty then B = empty.
3. Calculate mutual information between each pair of attributes base on equation (3).
4. Calculate threshold between each pair of attributes base on equation

$$\phi(X, Y) = \frac{I(X; Y)}{\sum_{i=1}^n T_{X,Y}} \text{ same as equation (4).}$$

$$m = \sum_{i=1}^n \text{no. frequency in all rows in the tables}$$

$$T_{X,Y} = \text{no. of variable in } X \times \text{no. of variable in } Y$$

5. Find the maximum of threshold = Max(ϕ).

6. Calculate weight for each attributes in the Table (i) base on equation (6).
7. Sorted all weight for all variables.
8. B = the maximum weight is parent and the other are evidences.
9. Calculate CPT for each node in structure B.
If nod = positive then

$$CPT = \left(\frac{\sum_{i=1}^m x = \text{yes}}{m} \right) \dots \dots \dots (7)$$
 Else

$$CPT = 1 - \left(\frac{\sum_{i=1}^m x = \text{no}}{m} \right) \dots \dots \dots (8)$$
 where, r is no. of row in the table
10. Calculate a probability for Table (i) base on equation

$$\text{Accum_Prob} = 1 - \left(\prod_{i=1}^n \text{Accum_Prob (node i)} \right) \dots \dots \dots (9)$$
End.

VII. EXPERIMENTAL RESULTS

The proposed system deals with two types of diseases heart and nervous system for experiment on the models in *MES*. Each table is stored testing disease cases. This was configured 10 tables, each table represents a particular disease five tables represent heart disease and the other five represent Diseases of the nervous system. As for the attributes for each table consists of a number of different attributes from the other table. Where the system has been tested by inserting symptoms and medical history of each patient. The enter data on the structure and calculates the probability of occurrence of the disease for each patient. So that each disease to a friendly structure of its own, this results show the likelihood of disease for each patient, and this is chosen top 2 of the possibilities of these diseases can make sure through the doctor examines exact for patient finds possibilities mentioned are correct and this is achieved the desired results, and this we find that the performance of the system is also in the table (3,4) 22 patients been taken every heart disease, as well as 20 patients been taken every disease of the nervous system. After that accuracy been calculated so we got to the results in every of the models in the system of *MES*. The performance of the *MES* can see in Tables (3, 4) for each disease. So that the rate of accuracy in the model of *Fast-CPT-Tree* is approximately (87.26%) in heart diseases and diseases of nervous system is approximately (91%), *FBC* is approximately (49.26%) in heart diseases and diseases of nervous system is approximately (75%) and *M-FBC* is approximately (93%) in heart diseases and diseases of nervous system is approximately (98%). This we can deduce that the rate of accuracy of the system *MES* for the model *M-FBC* is the best average results.

TABLE III. PERFORMANCE MES FOR HEART DISEASES .

Code Disease	Diseases Name	Number of Patients used in testing	Accuracy (%)		
			<i>Fast-CPT-Tree</i>	<i>FBC</i>	<i>M-FBC</i>
110	<i>High Blood Pressure</i>	22	81.8%	81.8%	100%
120	<i>Angina pectoris</i>	22	86.36%	36.36%	81.8%
121	<i>Myocardial Infarction</i>	22	90.9%	22.7%	95.45%
148	<i>Atrial Fibrillation</i>	22	90.9%	95.45%	100%
150	<i>Heart Failure</i>	22	86.36%	10%	86.36%
Average			87.26%	49.26%	93%

TABLE IV. PERFORMANCE MES FOR NERVOUS SYSTEM DISEASES.

Code Disease	Diseases Name	Number of Patients used in testing	Accuracy (%)		
			<i>Fast-CPT-Tree</i>	<i>FBC</i>	<i>M-FBC</i>
G40	<i>Epilepsy</i>	20	85%	85%	100%
G55	<i>Nerve Root and Plexus Compressions</i>	20	90%	80%	100%
G91	<i>Hydrocephalus</i>	20	95%	65%	100%
G94	<i>Head injury</i>	20	90%	65%	90%
G95	<i>Other diseases of spinal cord</i>	20	95%	80%	100%
Average			91%	75%	98%

This we can deduce that if many data were taken any of a large number of patients get a good result see Figure (4). This conclusion is based through access to the data in the Internet where all research is based on a particular disease either in this research is based on 10 diseases.

CONCLUSION

This paper has been presented the *M-FBC model*. It is depend on information theory by application mutual information on the attribute in table of diseases and graph by attribute converted to variable computation of weight for each variable and stored the variable form descending order and selected the maximum and put it the root for graph and put other variable as leaf for root or as child for the parent. By modified the equation for find CPT for each variable (node) and Probability for each disease (table). Where is tested the system by the doctors in the hospital in Iraq the accuracy for heart diseases are approximately (93%) and for nervous system diseases are approximately (98%). The experimental results appear that the M-FBC is best than on FBC when is tested the accuracy for heart diseases are approximately (49.26%) and for nervous system diseases are approximately (75%).

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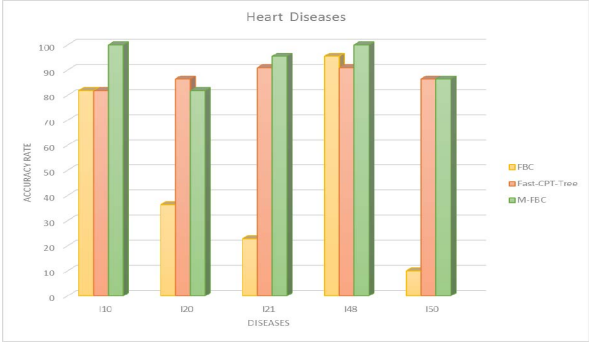
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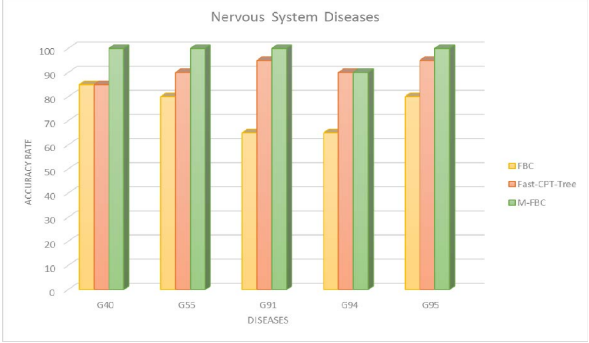
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(a)



(b)

Figure 4: (a) Performance MES for Heart Diseases, (b) Performance MES for Nervous System Diseases.